Working and Learning in Hard Times: Causal Evidence on Employment Downturns and Community College Students' Trajectories [*]

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Abstract

Community colleges enroll millions of low-income, working students who are uniquely exposed to local economic conditions. Despite evidence that aggregate CC enrollment is countercyclical, we lack causal evidence on how recessions affect students already enrolled. Using longitudinal administrative data from Minnesota (first-time CC enrollees, 2005-2012) linked to Unemployment Insurance wage records, this study employs a Bartik shift-share instrument embedded in a difference-in-differences design to identify the effects of local employment downturns on persistence, work behavior, and credential completion. Wild cluster bootstrap inference and Bonferroni correction address small cluster and multiple comparison concerns; event-study tests support parallel pre-trends. Preliminary results reveal substantial heterogeneity: recession exposure has no significant average effect, but Pell-eligible students and recently employed students persist more while full-time starters are significantly less likely to persist. Findings demonstrate how external economic shocks reshape trajectories for financially vulnerable, working learners.

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Introduction and Significance

Community colleges (CCs) enroll nearly half of all U.S. undergraduates, serving students who are disproportionately low-income, take fewer credits, work while enrolled, and are deeply embedded in local labor markets (NCES, 2022; Nichols & Valle, 2023). This local embeddedness amplifies students' vulnerability to employment downturns: most CC students work at some point during their enrollment, with working students averaging longer hours than four-year students (Velez et al., 2018); as commuter institutions serving geographically bound populations, CCs operate within regional economies that directly shape students' job opportunities (Reyes et al., 2019; Wu & McMillan, 2023; Ziskin et al., 2014).

CC students may face exacerbated financial and time constraints during economic disruptions. CCs often operate with fewer resources and student supports despite serving more students with high financial need (Baime & Baum, 2016). Particularly, students from lower-income backgrounds may work during college, prioritizing immediate income over career-relevant experience (Nichols & Valle, 2023) – two-thirds of working CC students hold jobs unrelated to their studies, including most studying in career and technical education (CTE) programs (NCES, 2020). Such misalignment between work and study has been shown to limit both college success and longer-term economic outcomes (Sanchez-Gelabert et al., 2017). While a robust literature establishes that CC enrollment is countercyclical – rising during economic downturns as opportunity costs of attending college decline (Barr & Turner, 2015; Foote & Grosz, 2020; Goodman & Winkelmann, 2025) – far less is known about how downturns affect the educational trajectories and labor market outcomes of students already enrolled in CCs.

Understanding how local employment downturns affect enrolled CC students' employment and enrollment trajectories, including work behaviors, program choices, and longer-term outcomes, is critical for multiple reasons. This research gap is particularly significant as CCs are increasingly tasked with aligning their programs to local labor market demands while serving as pathways to economic mobility for various student populations (Soliz, 2023), especially through CTE programs designed to prepare students for in-demand skilled occupations. Evidence on labor market scarring suggests that cohorts entering weak labor markets experience persistent employment and earnings penalties (Oreopoulos et al., 2012; Rothstein, 2023), raising the stakes of understanding CC students' college outcomes during downturns. Additionally, how students allocate time between employment and college coursework also remains empirically unexamined for CC students facing local labor market disruptions, despite being theoretically central to understanding educational investment decisions (Ost et al., 2018). As industries and communities experience recessions differently, CC students likely face varied pressures across both school and work (Ecton et al., 2023; Neyt et al., 2017), making their work-study trajectories during employment downturns consequential for educational and economic mobility.

The Great Recession (2008-2009) provides a consequential context for studying these dynamics as one of the largest employment downturns in modern U.S. history. Studying these historical patterns can inform how institutions and policymakers respond to today's economic volatility. Using statewide longitudinal administrative datasets linking CC students' enrollment and employment trajectories in Minnesota (2005-2013), this study addresses two research questions: 1) How do local employment downturns affect CC students' college outcomes: persistence, work patterns while enrolled, shift toward or away from CTE programs, credential

completion, and early-career earnings? 2) Do those effects vary across student subpopulations, and if so, how?

This study contributes to the literature on student employment, recession effects in higher education, and the role of CCs as countercyclical institutions, while offering methodological guidance on addressing the endogeneity of work behavior in causal designs. First, by treating local variation in economic recessions as a natural experiment, this study provides the first causal evidence on how aggregate labor market shocks affect the full population of already-enrolled CC students – reaching beyond those who are displaced or initially enroll during downturns – tracing the pathway from downturn exposure through work-behavior changes to completion and earnings. This fills a critical gap left by prior studies focused on individual-level displacement (e.g., Minaya et al., 2023; Ost et al., 2018), addressing a significant void in CC literature, as well as in research on adult and nontraditional students and in labor economics of education. Second, this work incorporates methodological innovations through integrating person-level administrative data, covering enrollment, work hours, and earnings, with county-level economic indicators constructed using a Bartik shift-share instrument (Bartik, 1991; Hershbein and Stuart, 2024; Stuart, 2022). This approach overcomes the common limitations of self-reported survey data while enabling the analysis of heterogeneous effects across student populations. Particularly, existing state-level administrative data systems rarely report hourly data on employment, with Minnesota being one of the few states reporting hours worked. This work therefore further reveals downturn-induced changes in work behavior among CC students and their influence on college outcomes, offering timely insights that could inform support for CC students in their dual roles as workers and learners during periods of economic uncertainty.

This research informs policy responses to economic volatility. The analysis provides actionable evidence for CCs' institutional planning during economic downturns: how to anticipate shifts in enrollment persistence, CTE program enrollment, work patterns, and subsequent completion and earnings outcomes that may influence the allocation of academic resources and program capacity. By examining impacts across industries and institutional contexts in Minnesota, the study provides evidence relevant to state initiatives targeting institutional resilience and workforce alignment, as well as to federal CTE funding decisions, during ongoing economic uncertainty.

The empirical strategy centers on a difference-in-differences (DiD) design that compares CC students across counties experiencing varying levels of employment declines to establish causality between economic conditions and student behaviors, leveraging county-level variation in downturn severity. The study employs a Bartik shift-share instrument that predicts county-level employment changes by interacting national industry trends with pre-recession local industry shares, thereby generating exogenous variation in local labor demand. For the second research question, this predicted employment downturns instruments for individual students' work hours and employment status within a two-stage least squares (2SLS) framework, yielding a Local Average Treatment Effect (LATE) of downturn-induced work intensity on longer-term outcomes. Heterogeneous effects across student characteristics (e.g., age, gender, economic background) and institutional contexts (urban vs. rural) reveal how different student populations adjust to local economic disruptions and where institutional or policy support may matter most.

Prior Research

How Community College Enrollment Responds to Industry Downturns

Recent evidence increasingly recognizes that college enrollment responses to labor market conditions vary by sector, with CCs exhibiting the strongest countercyclical response in the U.S. higher education system (Goodman and Winkelmann, 2025). Causal evidence consistently documents that CC enrollment rises during economic downturns (e.g., Foote & Grosz, 2020; Frenette et al., 2011; Goodman & Winkelmann, 2025; Barr & Turner, 2015). More recent studies further our understanding of the heterogeneous nature of this countercyclical response. In particular, downturn-induced CC enrollment gains appear to concentrate in vocational or CTE fields rather than academic or transfer-oriented programs (Foote & Grosz, 2020).[1] New enrollees' prior work experience also shapes these responses. For example, displaced workers from declining industries, such as manufacturing, are likelier to enroll in public colleges after job loss (Minaya et al., 2023), consistent with the idea that declining demand for industry-specific skills increases the relative attractiveness of retraining.

Recent high school graduates may also respond strategically to local downturns by shifting away from programs that are most directly related to the occupations impacted by recessions and substituting into related alternatives requiring similar skills, except in areas such as the skilled trades where few alternatives exist (Acton, 2021). Acton (2021) suggests that these responses vary by gender, program, and geography, with rural students appearing more responsive to local layoffs, likely due to the salience of such events in their information environment. Although

[1] While short-cycle CTE programs appear relatively "recession-proof" in community college research, studies at four-year colleges highlights STEM fields as more resilient (e.g., Ersoy, 2020), which may be informative for understanding transfer pathways, though the data do not allow for a full examination of transfer students' long-term completion by field.

these studies examine different populations, margins, and classifications of field or occupation, their findings can be complementary: Foote and Grosz's (2020) finding that manufacturing- and construction-related programs can grow after layoffs is compatible with Minaya's (2023) evidence that displaced manufacturing workers turn to college, and it is also compatible with Acton's (2021) finding that students may shift toward related alternatives when substitutes exist.

Nevertheless, the literature on countercyclical CC enrollment remains centered on first-time entry into college (e.g., Foote & Grosz, 2020; Goodman and Winkelmann, 2025), with limited evidence on what happens after CC students' initial enrollment, such as intensive margins of enrollment intensity and the subsequent behaviors. One important exception is Ost et al. (2018), pointing out that among working students, exposure to firm-level mass layoffs increases attempted credits but does not increase the probability of persistence – the intensive margin effect is concentrated among adult students age 25 and older, who likely face tighter time constraints.

Evidence on post-enrollment longer-term outcomes is thinner. Existing research shows the completion effects are generally minor, but they likely concentrate in shorter-term or high-return CTE fields. Foote and Grosz (2020), for instance, suggest mass layoffs increases CC credential completion primarily in short-term CTE programs rather than in associate degree programs, particularly in fields with strong labor market returns, such as allied health certificates and construction/manufacturing degrees – which complements Minaya et al.'s (2023) and Acton's (2021) evidence on short-term enrollment increases. In contrast, Goodman and Winkelmann (2025) and Stuart (2022) find no evidence on credential completion. Local economic fluctuations might offset the effects of college persistence and completion, as downturns can reduce the opportunity cost of schooling while simultaneously worsening

students' finances (Ost et al., 2018; Wu & McMillan, 2023). Cross-field heterogeneity deserves more attention, particularly whether students in CTE fields respond differently from those in other fields, as this informs CC program planning and resource allocation.

Direct evidence on earnings effects for CC students is particularly limited. Ost et al. (2018) show that in the short run mass layoffs reduce the probability of employment among students who were already working while enrolled. Regarding long-term effects, the *broader* recession literature points to persistent scarring in employment and earnings, but rarely for CC enrollees specifically. Hershbein and Stuart (2024) show that local labor market shocks generate persistent hysteresis across five recessions, with employment-population ratios remaining depressed and earnings losses concentrated in the lower and middle parts of the distribution. Stuart (2022), by contrast, studies exposure to the 1980-1982 recession during childhood and adolescence and finds the largest long-run earnings losses for those exposed at younger ages; that study is important for understanding long-run recession scarring, but it is not evidence about enrolled CC students. Related evidence on earnings losses at labor market entry comes largely from four-year or bachelor's-degree contexts (Oreopoulos et al., 2012; Rothstein, 2023).

At the same time, evidence on retraining suggests that CC education can yield earnings premiums after job displacement, providing important context for interpreting completion effects. Jacobson et al. (2005) show that one academic year of CC schooling yields long-term earnings effects for displaced workers, with particularly large returns for older women, even though older displaced workers participate less frequently than younger workers. Research suggests that institutions should be intentional in targeting resources and labor market information to students with diverse backgrounds and skill needs (Acton, 2021). Similarly, Hyman (2018) suggests that participants in federal Trade Adjustment Assistance programs

experienced higher earnings a decade after layoff, though gains diminished over time. Whether similar returns accrue to students who complete credentials during recessions, and whether those returns vary by the severity of local labor market conditions, remains underexplored.

The literature also varies considerably in how it defines and measures downturn exposure, and not all are strictly about the Great Recession. Some studies aggregate regional-level employment downturns and cannot track individuals over time (e.g., Foote & Grosz, 2020; Goodman and Winkelmann, 2025). Others analyze firm-level mass layoffs and their effects on individuals' enrollment choice after displacement (e.g., Minaya et al., 2023; Ost et al., 2018). Still others investigate occupation-specific aggregated downturns and how they affect program-specific enrollment (Acton, 2021). Studies of industry-specific downturns add important granularity by showing that local labor market disruptions are not uniform across sectors. For instance, manufacturing workers affected by trade shocks experience prolonged recovery periods and sustained earnings losses, though younger workers show greater mobility toward non-manufacturing sectors (Autor et al., 2025). Similarly, geographical variation may further the uneven effects. Descriptive evidence suggests that local economic conditions shape job availability and quality, thereby affecting CC students' employment prospects (Nichols & Valle, 2023; Reyes et al., 2019). Causal studies applying a Bartik-style shift-share instrument variable to exploit the fact that the recession was more severe in counties that specialized in certain industries support the geographical heterogeneity (e.g., Hershbein and Stuart, 2024; Stuart, 2022). This is especially relevant for Minnesota – for example, Central Minnesota's manufacturing and construction sectors experienced severe job losses while healthcare and educational services saw gains during the Great Recession (Greiner, 2025). Such uneven downturns likely altered the relative attractiveness of different programs and career pathways,

strengthening the case for examining heterogeneity across fields of study and local labor market contexts.

Simultaneous Enrollment and Employment: Response and Outcomes

The enrollment literature reviewed above treats work primarily as a background characteristic or an opportunity cost, rather than as a dynamic outcome that students adjust to during downturns. Yet for CC students – the majority of whom work at some point during enrollment – understanding how downturns reshape work behavior is essential to understanding educational trajectories. The second strand of literature examines these work-study trade-offs more directly.

Despite the prevalence of working students in CCs, research on how CC students navigate enrollment and employment simultaneously during economic shocks remains limited. Most recession studies focus on the general workforce, providing indirect evidence about students who study and work simultaneously. To my knowledge, Ost et al. (2018) provided the clearest causal evidence on this question in the US context. Using Ohio administrative data, they found that firm-level mass layoffs modestly increase attempted course credits among working students while significantly reducing their probability of employment, supporting the importance of reduced opportunity costs and time constraints in interpreting the educational response. This result is important because it suggests that downturns may change both students' labor supply and their academic behavior, even if persistence itself changes little.

Ost et al.'s (2018) study also highlights the endogeneity problem at the center of this study, although it does not fully resolve it. Their investigation of mass layoffs provides plausibly exogenous variation in employment for a subset of students, but this study examines a broader question of how aggregate local downturns affect the work-study choices of *all* CC students,

therefore requiring a different source of exogenous variation, which the Bartik-based design is intended to provide. To investigate the consequences of differentiating focal populations between this study and Ost et al's study, this study examines already-working students as one of the subgroup analyses. This way, we can better understand how and why already-working students might experience differently from *all* students.

More generally, evidence on how changes in work behaviors during college affect college outcomes remains limited, especially in CC settings and during recessions. I am not aware of prior work that explicitly estimates how downturn-induced changes in work behaviors affect general CC students' college outcomes using an identification strategy comparable to this study. Even outside the recession literature, much of the evidence on student employment comes from four-year institutions (Riggert et al., 2006), despite two-year students being overrepresented among working learners (Velez et al., 2018). Recent work using administrative data rather than self-reported data finds small average effects of working while enrolled on grades and other short-term academic outcomes (e.g., Darolia, 2014; Ecton et al., 2023). Darolia (2014) suggests that negative academic effects concentrate among full-time students, particularly in four-year settings, whereas part-time students and CC students appear not to be impacted academically. This pattern is important for CCs as diverse enrollment patterns and part-time enrollment are common (Crosta, 2014), suggesting the impact of working during college may vary significantly by enrollment intensity and work intensity.

Adverse consequences of working while enrolled appear larger for *persistence* and completion than for academic performance alone (Darolia, 2014; Neyt et al., 2017) or for post-college earnings (Douglas & Attewell, 2019; Witteveen & Attewell, 2021). The relationship between work and degree completion seems nonlinear, as negative associations becoming more

pronounced at higher weekly hours, with suggested thresholds ranging from around 10 hours (Ecton et al., 2023) to 20 hours (Soliz & Long, 2016). The limited causal studies on the impact of working during college still come from four-year colleges and generally suggest that financial aid has a limited impact on working students, although subsidized or academically aligned jobs may attenuate harms (Kim, 2022; Scott-Clayton & Minaya, 2016; Soliz & Long, 2016). For CC students, who tend to work more hours out of financial necessity, these patterns are particularly concerning. While jobs can bring essential financial support and valuable experiences, many students work in jobs unrelated to their studies (Nichols & Valle, 2023; Velez et al., 2018), which may ultimately constrain their workforce potential and human capital accumulation (Douglas & Attewell, 2019; Sanchez-Gelabert et al., 2017).

Summary: Gaps and Expected Contribution

This study contributes in several ways. First, it moves beyond the literature's emphasis on initial enrollment by examining the ongoing educational and employment choices of all CC enrollees. In doing so, it extends the analysis of CC responses to recessions from the extensive margin of college entry (e.g., Goodman & Winkelmann, 2025) to the intensive margins of persistence, course-taking intensity, and program choice. This is especially important because prior evidence either focuses on new entrants (e.g., Foote & Grosz, 2020) or on narrower subsets of enrolled students, such as those already working (Ost et al., 2018).

Second, it treats employment and enrollment as simultaneous, dynamic processes rather than as isolated outcomes. The analysis, therefore, examines a broader set of college outcomes that includes both educational trajectories and employment trajectories during and after enrollment.

Third, the dissertation addresses a methodological gap in CC studies. Existing quasi-experimental studies provide useful but incomplete evidence because most rely either on aggregate local shocks without individual longitudinal data or on individual shocks that do not capture broader local labor market conditions. This study directly connects region-level disruptions to individual educational and employment trajectories, identifying the causal effects of demand-side shocks on individual CC student trajectories with its DiD-Bartik design. Methodologically, this study is most similar to Hershbein and Stuart's (2024) and Stuart's (2022) for Bartik-based local recession exposure linked to long-run educational and income outcomes, although neither study focuses on CC students' enrollment-work trajectories. In terms of the focal population, this study is closest to Ost et al. (2018), although it extends beyond firm-level layoffs and students already employed at baseline.

Therefore, this work addresses these gaps by tracking longitudinal educational and employment trajectories of CC students through the Great Recession, beyond initial enrollment choice, while examining heterogeneity across student characteristics and institutional contexts.

Theoretical Frameworks

This study integrates two layers of theoretical perspectives – from individual to institutional levels – to examine how economic disruptions impact CC students' educational trajectories and college outcomes.

At the individual level, human capital theory provides a framework for understanding how external conditions shape students' educational investment decisions. Students make enrollment choices subject to multiple constraints – most notably, time and financial resources (Becker, 1962, 1965) – that often tighten during economic downturns. For CC students, who are

more likely to commute, work longer hours, and face financial vulnerability while having less institutional support, these constraints are especially pronounced. When local economies decline, students experience competing pressures: fewer local job opportunities reduce the opportunity cost of attending college but rising financial pressures and job insecurity simultaneously increase the need to earn income. Local labor market fluctuations, therefore, create offsetting forces that make their impact on college outcomes theoretically ambiguous. On the one hand, fewer employment opportunities can reduce the opportunity cost of college, allowing students to invest more time in their studies (Ost et al., 2018). On the other hand, labor market instabilities may worsen financial conditions, preventing students from affording college-related expenses (Sanchez-Gelabert et al., 2017). External financial sources, such as access to credit, may buffer the hardships caused by economic instability (Wu & McMillan, 2023). Costs of job search shape individuals' time constraints as well (Bhuller et al. 2023), which are not measured and separated from hours worked or academic credit load. Individuals' ability to adjust their work hours is itself is therefore constrained by employer flexibility and personal obligations.

The relationship between working while enrolled and college outcomes further complicates these mechanisms. Although the conventional view holds that working while enrolled harms academic outcomes (Becker, 1965; Riggert et al., 2006), work and study time are not simple one-for-one substitutes: reduced work hours do not necessarily translate into increased study time (Darolia, 2014), as students engage in complex time allocation decisions across multiple constraints and responsibilities (Kalenkoski & Pabilonia, 2012). Yet few studies treat employment and education as simultaneous, interdependent decisions (Ost et al., 2018). This gap is particularly consequential for CCs, where students more frequently rely on off-campus jobs and face tighter time and financial constraints than students at four-year institutions.

While the economic perspective emphasizes rational optimization under constraints, primary orientation theory highlights why students may respond differently to the same conditions (Baert et al., 2018). This behavioral theory posits that students' fundamental attitudes – whether they see themselves primarily as workers or as students – shape how they navigate competing demands. Those with strong work-oriented attitudes are more likely to experience negative academic effects from intensive employment, while those prioritizing academics may better maintain performance despite working (Baert et al., 2018; Mamiseishvili, 2010; Warren, 2002). These distinctions are particularly salient for CC students, who are more likely to be first-generation college students and from lower-income backgrounds, and who often have strong labor market attachment and extensive prior work experience – these students might be more likely to prioritize work over academics during economic downturns than their four-year peers. Although this study cannot directly observe individual orientations, students' education and employment backgrounds may serve as proxies for work orientation.

At the institutional level, organizational characteristics shape the constraints and opportunities students face when engaging with work and college. Institutions vary in selectivity and residential status, which in turn shape student populations' college/work orientations, financial resources (e.g., college costs, financial aid availability, job opportunities), academic requirements, and institutional practices that influence students' decision-making processes. These characteristics highlight why CCs deserve further attention in the empirical literature. CCs and other commuter, less-selective institutions, such as some regional universities (Ziskin et al., 2014), typically serve students who commute among home, college, and work. While these students may have lower living expenses and more flexible job options, they often face significant economic challenges (Nichols & Valle, 2023). In contrast, selective institutions with

high academic demands may require on-campus residence and restrict working hours, where students might receive more generous financial aid and better access to academically relevant on-campus jobs than those at commuter institutions. At the same time, students who do work at selective institutions may experience greater conflicts between work and study obligations, potentially hindering college success. Institutional practices are vital for students' college outcomes, particularly for nontraditional students (Remenick, 2019) and those from lower-income families who often rely on themselves to finance and complete their education (Rainey & Taylor, 2024).

These institutional differences are often nested within geographic regions (Hillman, 2025). Most college students attend nearby in-state colleges (Hillman, 2023), and CC students' educational and occupational opportunities are especially likely to be shaped by geography, including local employment circumstances and job prospects (Reyes et al., 2019). Studies recognize that job availability is a major determinant of CC students' decision-making (Reyes et al., 2019; Roessger et al., 2023), although some recent studies indicate that economic shocks affect both four-year and two-year institutions (Minaya et al., 2023; Wu & McMillan, 2023). Despite this importance, existing research has concentrated primarily on four-year universities, leaving significant knowledge gaps regarding working students at other institutional types.

Particularly, the variance in institutional capacity deserves particularly attention for the recession contexts, because of the surge in CC enrollments and shrinking in state budgets during downturns (Barr & Turner, 2013; Bound & Turner, 2007). At capacity-constrained colleges, continuing students might need to compete with new enrollees when persisting in colleges, including course seats, advising and related support, therefore more likely to be crowded out

after the initial enrollment. Most existing related literature on college cohort sizes and their effects on persistence and completion, however, concentrate on four-year college contexts.

Integrating these theoretical perspectives with the empirical evidence reviewed above yields several key empirical expectations, though the direction and magnitude of effects remain theoretically ambiguous due to offsetting forces. Regarding persistence and employment: Human capital theory predicts that downturns reduce the opportunity cost of schooling, thereby increasing persistence. However, if financial constraints bind more tightly than time constraints, students may be forced to exit college or increase work hours to meet basic needs. The net effect on persistence is therefore an empirical question (extensive margin), although recent evidence suggests continuing students may attempt more credits (intensive margin). Similarly, downturns should reduce employment probability on average (extensive margin), but the effect on hours worked conditional on employment (intensive margin) is ambiguous: students may reduce hours if time freed from work is reallocated to study, or they may increase hours if they face greater financial need or if remaining jobs require longer hours. The need to spend more time on job seeking might buffer the change in the intensive margin. Primary orientation theory suggests these responses may vary by students' work orientation (prioritizing work over other commitment, such as studying and caregiving), prior labor market experience, and financial aid status. The persistence effect and the employment effect at the extensive margin might therefore be concentrated among those who filed the FAFSA and are eligible for financial aid. Those who remain employed are more likely to remain in similar job sectors, particularly among those with stronger work orientation or from lower-income backgrounds.

Second, regarding program choice: If students persist and respond strategically to labor market signals, downturns may shift enrollment toward programs with stronger labor market

returns, such as certain sub-baccalaureate CTE programs (e.g., healthcare, construction) , especially among students with prior work experience in severely affected industries. However, this prediction assumes students have access to information about field-specific returns and the flexibility to switch programs, which may not hold for all students. The magnitude and heterogeneity of program-switching responses are therefore empirical questions.

Research Design and Method

Data and Sample

This study integrates multiple administrative datasets to examine how economic downturns affect the college and job trajectories of CC students. The primary data source is longitudinal administrative records from the Minnesota Office of Higher Education, covering student enrollment and demographics from 2005 to 2013. These records are linked to Minnesota's Unemployment Insurance (UI) system, providing uniquely comprehensive employment tracking, including quarterly earnings, worked hours, and industry sectors classified under the North American Industry Classification System (NAICS). This UI linkage offers significant advantages over self-reported survey data, which often suffers from response bias (Sanchez-Gelabert et al., 2017) or requires imputing wages from limited information (Ecton et al., 2023), thereby enabling precise measurement of work intensity and job characteristics. The study supplements these individual-level data with county-level employment statistics by industry from the Quarterly Census of Employment and Wages (QCEW), which uses the same industry classification system. Institutional characteristics come from the Integrated Postsecondary Education Data System.

Minnesota provides an ideal research setting for several reasons. First, the state's economic and industrial diversity, spanning retail, manufacturing, agriculture, services, and technology, creates meaningful variation in recession exposure across counties. Second, the metro-rural contrast generates variation in both economic and institutional contexts. Third, Minnesota's robust data infrastructure enables precise measurement of the key relationships under study. Finally, the state's CC system serves diverse populations across varying economic contexts, enhancing external validity.

The initial sample comprises approximately 355,000 first-time CC enrollees from 2005 to 2012 academic years, observed across 35 institutional counties over nine cohort years, spanning pre-recession, recession (2008-2009), and post-recession periods. The panel structure tracks students' year-by-year enrollment and employment decisions, enabling analysis of dynamic responses to economic downturns. Table 1 presents the descriptive characteristics for CC students in this sample.

Table 1

Sample Characteristics

	Full Sample	Entry Cohort	
		Post-Recession	Pre-Recession
N	347,103	188,826	158,277
Share of Full Sample (%)	100%	54%	46%
Demographic Characteristics			
Age at Entry (mean)	27	28	27
Aged 25 or Older (%)	44%	44%	44%
Female (%)	50%	49%	52%
White (%)	75%	71%	79%
Black/African-American (%)	11%	13%	10%
Hispanic/Latino (%)	4%	5%	3%
Asian (%)	5%	5%	4%
Native American/Pacific Islander (%)	2%	2%	2%
Other/Unknown Race/Ethnicity (%)	3%	4%	2%
Enrollment Characteristics			
Full-Time Starter (%)	58%	58%	58%
Enrolled Credit Hours (mean)	11	11	11
Employment History			
Worked Two Years Before Entry (%)	63%	60%	68%
Worked One Year Before Entry (%)	71%	67%	76%
Worked at Entry (%)	74%	70%	79%
Weekly Hours Worked while Enrolled, Unconditional (mean)	15	13	17
Financial Characteristics			
Submitted FAFSA (%)*	72%	83%	60%
Pell Eligible (%)*	67%	72%	58%
First-Generation College Student (%)*	42%	43%	40%
Institutional Characteristics			
Public Two-Year Colleges (%)	100%	100%	100%
Rural Colleges (%)	9%	8%	9%
Small Two-Year Colleges (%)	16%	16%	16%
Large Two-Year Colleges (%)	16%	16%	16%

Notes. This table summarizes characteristics of sampled students at the year of their first entry in a Minnesota community college. Due to small sample sizes, students identifying as Native American and those identifying as Pacific Islander were combined into a single Native American / Pacific Islander category to ensure statistical stability. Multiracial and Nonresident students are combined into a single group of Other/Unknown Race/Ethnicity for the same reason. * Pell-eligible and First-generation status are conditional on students submitted FAFSA; whether enrolled in CTE program is conditional on major data availability.

Measures

Key Outcomes

The study examines students' behavioral responses along persistence and earnings. Each dimension focuses on one pre-specified primary outcome.

College persistence focuses on year-to-year enrollment persistence (persisted or not) up to four years after enrollment. Secondary outcomes measure the intensity of college persistence (persisted full-time or not). Employment during enrollment is also examined as secondary outcomes, including employment status and work intensity while persisting. Although full results are not presented in this paper, my full analysis includes secondary outcomes on program choice (CTE or not) and major switching.

Relatedly, college credential data is linked but available only for tracking two years for post-recession cohorts; therefore, such early completion is estimated as secondary outcomes for supplementary analysis. Early credential completion is operationalized as a cumulative annual indicator observed each year following initial enrollment, equal to one in the year a credential is earned and all subsequent years, and zero otherwise. This structure allows the DiD model to estimate both whether students completed credentials and when, capturing acceleration or delays in completion trajectories.

Annual earnings up to three years after college entry are observed as primary outcomes. Unconditional earnings measure includes students with zero annual earnings, capturing the combined effect of employment downturn exposure on employment and wages. Hourly wage supplements the analysis with wage level information. Secondary outcomes also measure students' employment status while enrolled as well as conditional earnings to isolate the wage effect among employed workers. Notably, the working-related behavioral responses are

conditional on the individual holding valid employment records in the UI data, they cannot track out-of-state employment records. Nevertheless, examining their in-state earning trajectories provides a more comprehensive picture of how economic shocks reshape working students' educational and employment trajectories.

Treatment: Employment Downturn

To measure how the Great Recession affected local job markets, the study uses county-level employment data categorized by industry from the QCEW data. Measuring the impact of a recession is methodologically challenging: relying only on observed changes in local employment rates can be misleading, since local students' work and enrollment behaviors may themselves influence those rates, creating a risk of circular causality.

To address this, the study employs a well-established economic approach called the Bartik-style shift-share instrument (Autor & Dorn, 2013; Goldsmith-Pinkham et al., 2020) to ensure that downturns are driven by national industry trends rather than local factors that might correlate with student characteristics. Following Stuart's (2022) approach of Bartik-style predicted local changes, this study creates a predicted measure of local economic downturns by combining each county's pre-recession industry composition with national trends in those industries during 2008-2009. The logic is straightforward: counties with larger shares of jobs in nationally declining industries (e.g., manufacturing) should experience more severe predicted downturns, while counties with jobs in stable industries (e.g., healthcare) should be relatively protected. The predicted employment downturn is:

$$Downturn_C = \sum_j (Share_{cj,2007} \times NationalChange_{j,2008-2009}) \quad (1)$$

where $Share_{cj,2007}$ represents industry *j*'s share of county *c*'s employment in 2007 (pre-recession), and $NationalChange_{j,2008-2009}$ captures the national employment change in

industry *j* during the Great Recession. The study validates this measure using two core conditions for causal identification: first, testing whether predicted downturns correlate with actual local changes (relevance); second, confirming that national trends affect student behaviors only through local economic downturns, not directly (exclusion restriction), supported by prior research documenting minimal direct effects of national trends on local behaviors (Autor & Dorn, 2013; Goldsmith-Pinkham et al., 2020).

Continuous vs. Binary Treatment. When operationalizing regional downturns, prior literature mostly uses continuous treatment as the primary measure (e.g., Goodman and Winkelmann, 2025; Stuart, 2022), and some use binary measures that compare counties above and below the median decrease as secondary measures (e.g., Stuart, 2022). Following Stuart's (2022) and Hershbein and Stuart's (2024) approach of constructing a continuous primary measure and a secondary binary measure, the main approach of this study has been constructing continuous measures for downturns that hit in 2008, while the study also introduces binary treatment variables as a robustness check. Binary treatment requires grouping counties based on their experienced downturns, capturing the extensive margins of the aggregated downturns. Continuous measures provide greater granularity on the impact of downturns.

Static vs. Dynamic Treatment. The study measures county industry composition prior to the onset of the recession and combines it with national industry employment changes to construct both static and dynamic annual Bartik measures of local labor demand shocks. Static Bartik measures of industry downturns during the 2008-2009 period are the primary focus, as they directly capture the Great Recession, while the dynamic Bartik measure uses each county's peak-to-trough employment decline, which allows for variation in local recession severity and timing and provides supplemental insights into adjustment dynamics. Notably, the 2008

recession was a common aggregate shock, as all counties experienced it around the same time but to varying degrees. This makes *static DiD* the natural primary specification (Equation 2), following Stuart's (2022) approach. In comparison, *dynamic DiD*, as in Hershbein and Stuart's (2024), is more appropriate when treatment genuinely rolls out at different times and the main focus is to test whether effects vary by exposure length (Callaway & Sant'Anna, 2021). The study aims to conduct dynamic DiD with continuous treatment measures as an extension and sensitivity checks.

Analytical Approaches

The core identification strategy leverages DiD estimation (Baker et al., 2025; Roth et al., 2023), which compares changes in student outcomes across counties that were differentially exposed to the Great Recession before and after 2008. Equation (2) examines the direct impact of employment downturns on students' work intensity and academic focus:

$$Y_{ict} = \alpha_0 + \alpha_1(Downturn_c \times Post_t) + \gamma X_{ict} + \lambda_c + \mu_t + \varepsilon_{ict} \quad (2)$$

where Y_{ict} indicates each outcome for student *i* in county *c* in year *t*. The main variable of interest, $(Downturn_c \times Post_t)$, captures the drop in employment associated with the national industry trends during the Great Recession. $Downturn_c$ is the predicted employment decline during the recession period from Equation 1, while $Post_t$ represents the post-recession period (2008 onward). The coefficient α_1 captures how student behavior shifted per unit of downturn intensity – i.e., whether students worked more/less or enrolled differently in response to an economic recession. This study also examines whether these behavioral responses vary by student characteristics (age at entry, gender, race/ethnicity, first-generation status, Pell eligibility, work history, and enrollment intensity) and institutional contexts (urban/suburban/rural settings, institutional size) by including triple interaction terms. The study further examines policy-

relevant populations, particularly adult learners aged 25 and older in CCs, who may be most affected by economic disruptions. The model includes county and year fixed effects (λ_c, μ_t) to account for unobserved differences across counties and time to improve the reliability of the estimated effect.

This study employs multiple strategies to address threats to causal inference. First, the study verifies the core assumption underlying the DiD approach through event-study analysis that tracks outcomes over time relative to recession onset, ensuring that counties with different degrees of employment downturns followed similar pre-recession trajectories (the parallel trends assumption).[2] Second, the study tests whether the findings depend on specific modeling assumptions by sensitivity checks, examining alternative statistical specifications, including flexible spline functions and categorical work hours measures, to ensure results are not driven by restrictive functional forms.

Heterogeneity Analysis

Heterogeneity analysis examines potential subpopulation differences in the outcomes studied, using triple-interactions between the indicator for the focal subpopulation and the main variable of interest within the DiD models. As supplementary analysis, the main models are also estimated on subsamples to get straightforward estimates. With counties being limited within one state's CC student sample, estimates from the interaction terms are primary subgroup estimates.

I focus on heterogeneous effects across student background characteristics to uncover how different student populations adjust to local economic disruptions. Institutional-level and

[2] This examination of pre-recession trends is essential for establishing whether observed post-recession differences can be attributed to the economic downturn rather than pre-existing disparities. Should this assumption be violated, the study will implement covariate balancing propensity score methods within the DiD framework, which provides additional protection against bias by ensuring treatment and comparison groups are more similar on observable characteristics (Li & Matsushita, 2025).

county-level heterogeneity are examined as exploratory factors, due to the limited power in this county-level analysis. Any evidence on institutional and county subgroups are suggestive and should not be interpreted as causal estimates.

Student Characteristics (Primary). This study examines student-level heterogeneous effects across five sets of background characteristics: adult student status (age at entry>25), initial enrollment status (full-time at entry), pre-college employment status, financial background (Pell-eligible and FAFSA-filing status), and educational background (first-generation status). These characteristics capture meaningful differences in students' labor market attachment, financial circumstances, and academic engagement patterns that may moderate responses to local employment downturns.

Age at entry is among the most consequential characteristics for understanding CC students' trajectories and outcomes during recessions. This variable is constructed as a binary divided by whether a student reaches 25 years old at college entry. Prior evidence suggests adult students, defined as those 25 and older, may experience concentrated gains in course credits attempted and grades performance following mass layoffs (Ost et al., 2018), consistent with the view that older students have stronger labor market attachment and face more immediate trade-offs between work and schooling

Enrollment intensity at entry is operationalized through full-time versus part-time enrollment at the entry point in CCs. This division likely shapes students' trajectories and the subsequent outcomes, as part-time enrollment is common in CCs (Crosta, 2014), and prior research suggests the impact of working while enrolled might be weak among part-time students (Darolia, 2014).

This study examines baseline employment status as an importance source of heterogeneity, measured twice before students' college entry (two years and one year pre-college, respectively). Because one of this study's stated contributions is its broader scope relative to Ost et al. (2018), which focuses on the impact on working students, it would be analytically meaningful to examine whether the persistence and enrollment effects play out differently for students who were working during college versus those who were not. Additionally, pre-enrollment work history – whether students were employed prior to CC enrollment – is examined as another proxy for work orientation, with students who have more stable pre-enrollment work history expected to have stronger labor market attachment and work-oriented attitudes and therefore might prioritize employment over education despite during downturns; students who were already working might also experience more direct disruption from local employment shocks. Therefore, I examine students' employment status at multiple time points: whether students worked two years prior to CC entry and one year prior to CC entry are the primary focus.

Financial backgrounds include two proxies: Pell grant eligibility and FAFSA filing status. Research suggests lower-earning groups are significantly more likely to enroll in CC than higher earners following mass layoffs (Minaya et al., 2023), and the earnings decline tend to concentrate in the bottom and middle of the wage distribution (Hershbein and Stuart, 2024), making economically disadvantaged students a theoretically motivated subgroup. Pell eligibility is a useful indicator of low-income background, though an imperfect one (Delisle 2017), and it is observed only for students who file a FAFSA. That conditionality is not purely a nuisance. Filing reflects both intention and access to help navigating the process: assistance with completing the form raises submission rates, while information alone does not (Bettinger et al. 2012; Dynarski

& Scott-Clayton, 2006). Filing status therefore carries information of its own, and it also determines whether Pell eligibility and first-generation status are observed at all.

Educational background is captured through first-generation college student status, conditional on FAFSA submission. First-generation students face distinct financial pressures and weaker institutional integration than their continuing-generation peers, which may amplify both their responsiveness to labor market disruptions and their vulnerability to stop-out (Mamiseishvili, 2010). This measure also serves as a proxy for work orientation, as first-generation students may exhibit stronger work-oriented attitudes.

Gender and race/ethnicity are examined in the descriptive analysis. While prior studies suggest limited evidence of gender or race/ethnicity differences in enrollment or persistence responses to downturns (Foote & Grosz, 2020; Goodman and Winkelmann, 2025; Ost et al., 2018), gender gaps might be valid for industry-specific downturns (Acton, 2021). Race/ethnicity groups include Black/African-American, Hispanic/Latino, Asian, Native American/Pacific Islander, and Others.[3] These characteristics are included as descriptive checks given their established relevance to CC outcomes more broadly and potential interactions with the industry-specific nature of the employment downturns examined here.

Institutional and County Characteristics (Exploratory). The institutional subgroup analysis focuses on rural versus urban classification, drawing on IPEDS data. Prior literature suggests that rural students drive CC enrollment responses to mass layoffs more strongly than urban students, potentially reflecting stronger geographic preferences that make local labor

[3] Due to small sample sizes, students identifying as Native American and those identifying as Pacific Islander were combined into a single Native American / Pacific Islander category to ensure statistical stability. Multiracial and Nonresident students are combined into a single group of Other/Unknown Race/Ethnicity for the same reason.

market disruptions more salient (Acton, 2021), informing the urban-rural distinction as a theoretically motivated comparison. Further, local industry composition might shape CC students' enrollment and employment choices as well. Based on the industry prevalence in Minnesota, three industry-concentration subgroups are examined: counties with below-median manufacturing employment share, below-median retail employment share, and below-median healthcare employment share. Students in a manufacturing-heavy county likely behave differently during employment downturns compared to a healthcare-heavy county, both because the types of jobs available differ and because the CTE programs at local colleges are designed around the local industry mix. Nevertheless, these classification types further limit the county-level variation and the power of the analysis; any evidence on these contextual subgroups is suggestive and should not be interpreted as causal.

Robustness Checks

When the number of county clusters is low or the size of counties vary greatly in the sample, conventional cluster-robust variance estimators may be over-confident (Cameron et al., 2008; Roodman et al., 2019). The wild cluster bootstrap (WCB) corrects for this by imposing the null hypothesis and resampling cluster-level residuals. All main findings reported as statistically significant above are validated by the WCB.

Additionally, this study examines seven main subgroups, the overall Family-Wise Error Rate (FWER) is inflated. To account for these multiple comparisons, I apply a Bonferroni correction within each outcome across the seven subgroup tests ($\alpha^* = \alpha/7 = 0.007$). Consequently, only findings with a WCB p-value below 0.007 are considered statistically significant after adjustment. This approach provides a highly strict and conservative control over the FWER.

Research Settings: Education and Economic Context in Minnesota.

Minnesota's counties differ sharply in industry composition, which supports credible identification and generates variation in the shift-share exposure measure. While the Minneapolis-St. Paul metropolitan area exhibits strong labor demand in health care, finance, and business management, non-metropolitan counties demonstrate substantial employment concentrations in durable goods manufacturing, agriculture, and mining along the Iron Range: Northeast Minnesota in mining, with transportation equipment manufacturing around Duluth; Southwest Minnesota in agriculture and food processing; Southeast Minnesota in manufacturing, including computer and electronic products (Minnesota Chamber Foundation, 2024).

Minnesota's educational and training context during the studied period also deserves additional attention. In Minnesota, public two-year postsecondary education, i.e., CC education, is governed under the unified Minnesota State Colleges and Universities system, with approximately 30 CCs operating under centralized tuition-setting and credit-transfer frameworks. Like other states, CTE programs take up significant portion of Minnesota's CC offerings.[4]

Minnesota's CC CTE offerings track regional industry structure, clustering in a few primary industry domains: healthcare (e.g., nursing, allied health), manufacturing (e.g., mechanics and repair technologies; precision production, including welding), and business and

[4] Following federal Perkins standards and state administrative classifications, postsecondary Career and Technical Education (CTE) programs are defined as credit-bearing curricula organized around specific occupational credentials, primarily on the sub-baccalaureate levels, such as associate degrees and certificates in business, healthcare, and manufacturing (NCES, n.d.-a). These programs are designed for direct labor market entry, while also opening doors for transfer toward higher-level degrees. The NCES Postsecondary Taxonomy encompasses 26 fields of study at the level of two-digit CIP codes: Agriculture (01), Natural resources (03), Business and marketing (52), Communication (09), Communications technologies (10), Computer and information sciences (11), Family and consumer sciences (19), Personal and culinary services (12), Parks, recreation, and fitness (31), Education (13), Architecture (04), Engineering (14), Engineering technologies (15), Military technologies (29), Science technologies (41), Health sciences (51), Protective services (43), Military services (28), Legal professions and studies (22), Library science (25), Public administration and social services (44), Theology and religious vocations (39), Construction (46), Repair (47), Manufacturing (48), and Transportation (49) (NCES, n.d.-b).

marketing (Minnesota State Colleges and Universities, 2025). Other fields, including construction, transportation, and agricultural mechanics, account for smaller shares statewide but concentrating in certain regions. Consequently, program availability mirrors baseline county economic structures, creating distinct regional pathways for displaced workers and new entrants. This context necessitates the analysis on regional level heterogeneity of the economies.

Notably, a critical policy change following the Great Recession was federal workforce training investments, notably the Trade Adjustment Assistance Community College and Career Training (TAACCCT) grant program administered by the U.S. Department of Labor between 2011 and 2014 (U.S. Department of Labor, n.d.; Blume et al., 2019). Minnesota institutions secured major TAACCCT consortium and individual grants through multi-institution and regional partnerships targeting high-demand occupational pathways, primarily through advanced manufacturing workforce redesign and healthcare initiatives.[5] These federal grants funded targeted curriculum redesigns, embedded industry-recognized credentials, expanded credit-for-prior-learning assessments, and established modularized short-term certificate pathways in precision machining and health informatics. Because TAACCCT awards rolled out in the later

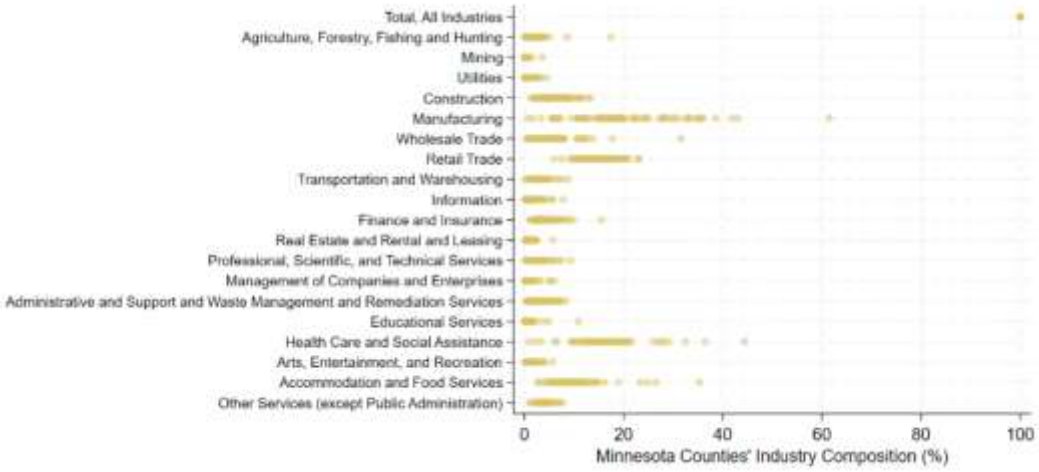
[5] Over the four rounds of the TAACCCT grant program, Minnesota institutions participated in seven major grant projects: 1. Minnesota Advanced Manufacturing Program, led by South Central College (Mankato, MN). An expansive 12-college state consortium on manufacturing skills using apprenticeship frameworks. 2. Advanced Manufacturing Education Alliance, led by Central Lakes College (Brainerd, MN). A multi-college regional network focused on building free digital curriculum for precision tools, welding, and remote video classrooms. 3. Rural Information Technology Alliance, led by Pine Technical College (Pine City, MN). A 4-college regional consortium working alongside rural workforce boards to fast-track adults into tech-support and network careers. 4. Open-Access Orthotics and Prosthetics Education Network, led by Century College (White Bear Lake, MN). A nationwide, 5-college health services grant expanding credential access to combat shortages in clinical medical tech manufacturing. 5. Unmanned Aircraft Systems and Geospatial Information Technology Integration, led by Northland Community and Technical College (Thief River Falls, MN). An individual program grant designed to build advanced certificate pathways for imagery analysis and aerial drone maintenance. 6. 360° Manufacturing and Applied Engineering Regional Center of Excellence, led by Bemidji State University (Bemidji, MN). A targeted multi-tier program building modular engineering curriculum that seamlessly connects regional manufacturing employers to local classrooms. 7. Mississippi River Transportation, Distribution, and Logistics Consortium, led by Lewis and Clark Community College (Godfrey, IL), with Minnesota State College Southeast (Winona, MN) serving as the main Minnesota operational lead hub. A massive multi-state consortium involving nine colleges along the Mississippi River corridor to train workers for supply chain, warehousing, and commercial transport roles.

cohorts of the sample window (2011-2012), they may have lowered the marginal cost of selecting technical certificates in hard-hit industrial counties.

The distribution of industries across counties makes the identification strategy work within boundaries. Because Minnesota's downturns have historically been shallower than the nation's, the milder aggregate environment also limits the scope for endogenous migration or institutional closure to confound county-level exposure. During the Great Recession, state unemployment peaked at 8.5 percent against a national peak of 10 percent, and state output had returned to its pre-recession level by 2010 while national output had not (Johnston, 2010; Phelan, 2020). Statewide, with healthcare and retail, two of the largest industries in the state (see Figure 1), remaining resilient during economic recessions, Minnesota was relatively protected from national economic downturns during the Great Recession, although each industry's employment downturns are slightly lower than the national average (Figure 2).

Figure 1

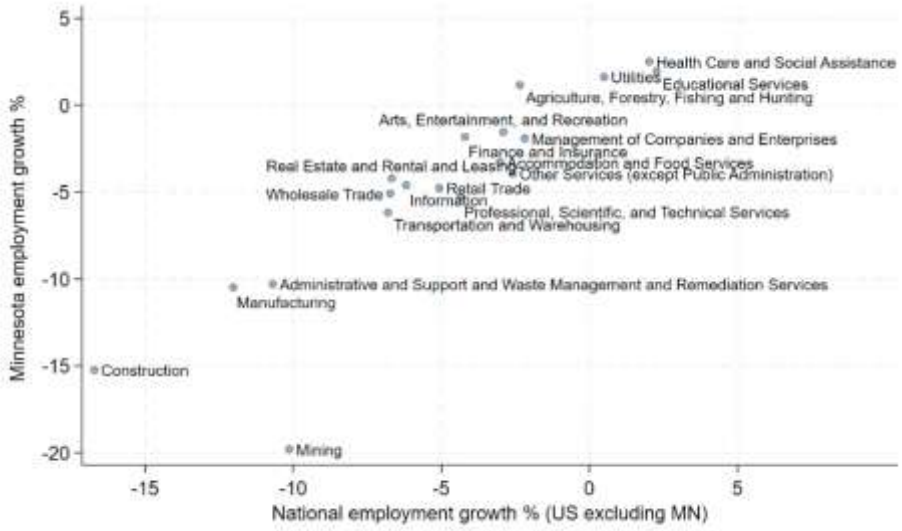
Industry Composition of Minnesota Counties in 2007



Note. Industry sectors are measured by the NAICS (2-digit level). Employment data come from the private sectors to align with available data on the national level in QCEW.

Figure 2

Employment Downturns during the 2008-2009 Great Recession (%) by Industry Sectors: Minnesota vs National (US Excluding MN)



Note. Employment downturn measures the percentage changes in the industry during the Great Recession (from 2008 to 2009). Industry sectors are measured by the NAICS (2-digit level). Employment data come from the private sectors to align with available data on the national level in QCEW.

Nevertheless, the relevant variation for identification is supported cross-county, which is presented as part of the descriptive patterns of employment downturns.

Preliminary Findings

Descriptive Patterns

The initial analyses establish the feasibility of this study and provide encouraging preliminary evidence. Descriptive statistics confirm substantial variation in local employment downturns across Minnesota counties during the Great Recession, with sufficient heterogeneity to detect differential impacts on student outcomes. The distribution of Bartik and realized employment declines across all 87 Minnesotan counties is presented in Appendix A (Figure A1). Predicted employment declines range from around 15% in recession-resistant counties with healthcare concentration (e.g., Olmsted County, home to Mayo Clinic) to approximately 20~30%

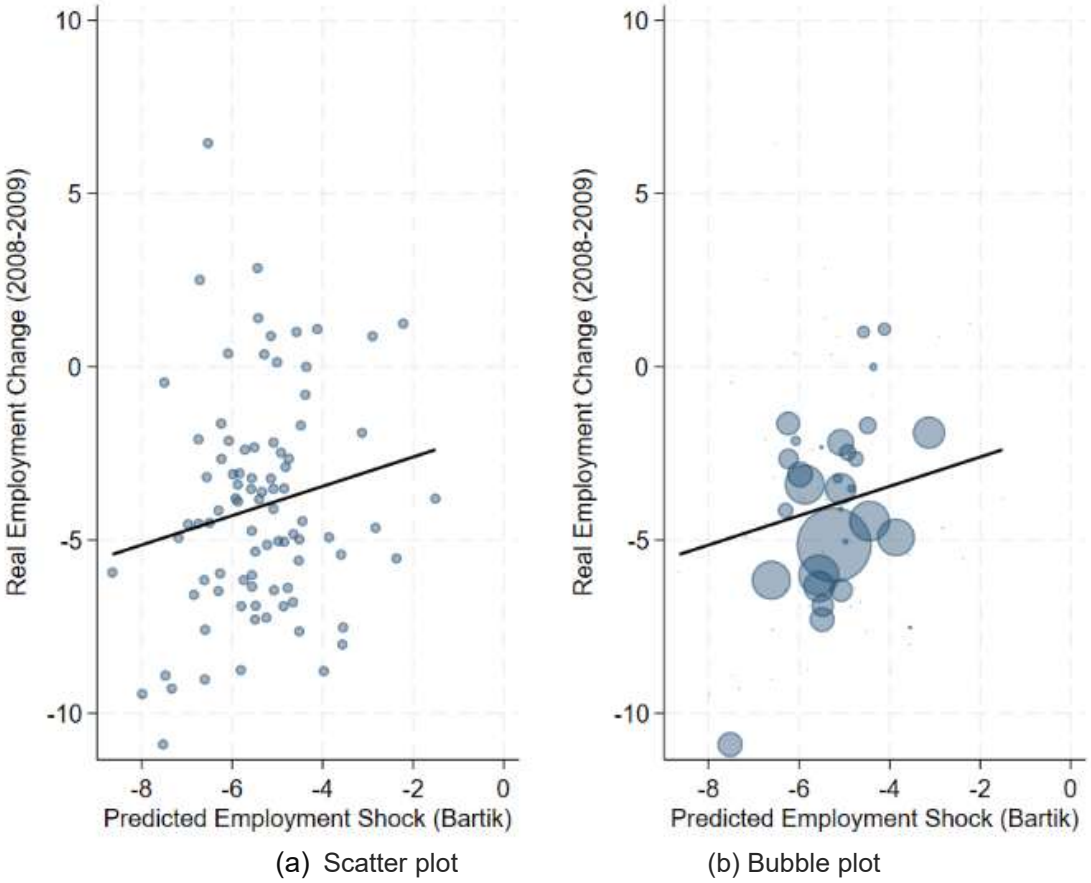
in manufacturing-dependent counties (e.g., Anoka County, Nobles County). Largest counties present include Hennepin (largest), Ramsey, Dakota, Anoka, and Washington counties.

Figure 3 presents a pair of comparisons between realized employment changes and estimated Bartik employment exposure during the Great Recession. Figure 3(a) compares the Bartik measure with realized county employment changes during the Great Recession for all 87 counties in Minnesota. The relationship between the Bartik-style change and the realized change is positive but modest ($\rho = 0.424$), consistent with the Bartik measure capturing ex ante exposure to national industry shocks rather than realized employment changes. Realized employment changes additionally reflect county-specific adjustment, migration, local demand conditions, and other idiosyncratic shocks. The Bartik measure is not intended to reproduce realized county employment changes. Instead, it captures predetermined exposure to national industry shocks based on pre-recession industrial composition. The modest correlation is therefore expected because realized employment outcomes additionally reflect local adjustment mechanisms and idiosyncratic county-specific shocks.

The analytical sample is not systematically different in terms of the Bartik-realized shock relationship. Built upon Figure 3(a), Figure 3(b) plots the relationship for the counties containing sample institutions , where bubble sizes scale with student enrollment. Although students originate from every county in Minnesota, enrollments occur almost exclusively in colleges located in 35 counties. Consequently, local labor market conditions are measured at the county of the enrolled institution, resulting in treatment variation across 35 institutional counties. Figure 3(a) and Figure 3(b) suggest that institution counties are not unusual relative to the rest of Minnesota.

Figure 3

Realized Employment Change vs Bartik Employment Exposure (%), Minnesota Counties, 2008-2009



Notes. (a) all 87 MN counties; (b) bubble size = number of students in the analytical sample enrolled in that county's institutions. Bartik measure constructed from 2007 county industry shares × national industry employment change, 2008-2009. Downturn measures are expressed as their original negative values (i.e., proportional employment declines during the 2008-2009 recession) to preserve intuitive directionality.

Because inference is clustered at the county level and the number of clusters is modest, inference relies on wild cluster bootstrapping (Cameron et al., 2008; Roodman et al., 2019), reweighting residuals at the cluster level, which preserves the cluster structure and works with few clusters. I report p-values from the wild cluster bootstrap using the restricted wild-cluster

bootstrap implemented in Roodman et al.'s *boottest*.[6] I use Rademacher weights by default and Webb weights in sensitivity checks.

The study constructs a binary measure of downturns to better present the trends of student behaviors across counties with different recession severity, dichotomizing counties by the median level of predicted downturns. Additionally, this analysis also examines dosage effects by constructing the continuous measure of county-level downturns. As shown in Appendix Figure A2, those experiencing severe downturn are constructed as treated groups, and the others are comparison groups, respectively. Students from counties that experienced a steeper downturn tend to enroll full-time and work less than the comparison groups; although there is no visual difference in the share of students enrolled in CTE programs, CTE majors increase steadily during the recession. These descriptive trends align with the prior literature of countercyclical CC enrollment. Further, heavier-hit counties show slightly lower work-academic alignment than their peers in most of the selected fields, except for health, which is the largest group with non-missing values in the work-academic alignment variable, indicating the descriptive gaps might be a consequence of limited subsample size, which deserves further analysis.

Difference-in-Differences Estimation

In general, no evidence of differential pre-trends is found, with event-study estimates and joint tests support the parallel-trends assumption. Event-study analyses show stable pre-recession trends in work hours and enrollment patterns, with clear divergence beginning in 2009, providing

[6] For full sample analyses, inference is driven by the classic Wild Cluster Bootstrap (Wald-type). For highly parameterized interaction models and reduced subgroup samples, I utilize the Score Bootstrap and Webb weights respectively to guarantee model convergence and accommodate smaller cluster environments.

initial evidence consistent with recession effects on student behavior. Divergence is consistent across subpopulations, although its extent varies.

This subsection focuses on the difference-in-differences estimates. In the full sample, students in harder-hit counties show a modest positive point estimate for next-year enrollment persistence (+2.1 percentage points, or 2.1pp), which is not statistically significant (WCB p = 0.108). The continuous specification yields a near-zero estimate (WCB p = 0.519). Aggregate outcomes are not detectably different in shock counties versus non-shock counties. These null finding at the aggregate conceal substantial, theoretically meaningful heterogeneity, which is dissected below.

The early completion results, despite it is measured short-term and cannot represent ultimate completion status, present an opposite pattern. Above-median employment downturn is associated with a 5.8 percentage points reduction in annual credential attainment in the full sample. Its magnitude is striking, a roughly 26 percent relative decline from a baseline rate of approximately 22 percent.

Persistence

Table 2 presents the estimates on persistence outcome. Panel A includes the binary TWFE estimate of how the harder-hit county impact student outcomes and Panel B is the continuous Bartik DiD estimate of how every one percent of employment downturn affects student outcomes. To estimate subgroup effects, both interacted with an indicator for the focal subpopulation.

Unlike the null aggregated pattern, the recession effects on college persistence show considerable variation across student subpopulation, among which the most striking finding is the persistence of Pell-eligible students (Table 3). In the interaction model on the binary measure

of local downturns, Pell-eligible students show a differential persistence rate 17.5 percentage points (pp) higher than non-Pell students in similarly impacted counties (WCB $p < 0.007$). This estimate survives every correction applied with conventional significance, wild cluster bootstrap (WCB $p \approx 0.000$), and the Bonferroni threshold ($\alpha^* = 0.007$). Relatedly, the FAFSA-filers result shows a qualitatively similar pattern in the persistence advantage (+8.4 pp, WCB $p < 0.007$) that misses the Bonferroni threshold, indicating a suggestive pattern. [7]

Recently employed students, who worked in the one or two years immediately before enrolling, also show persistence advantages in shock counties, suggesting notable reaction to labor market contractions. Those who worked two years before enrolling are 2.9 percentage points more likely to persist (WCB $p < 0.007$), which is slightly smaller than the probably for those who worked one year pre-college (3.9 pp, WCB $p < 0.007$), both surviving the Bonferroni threshold.

In contrast, full-time starters show a significant yet opposite pattern in persistence outcome, 7.8 percentage points *less* likely to persist in counties experiencing above-median employment losses (WCB $p < 0.007$). Adult students show no differential effect as its 0.7 percentage-point lower result has a WCB p-value of 0.559, which is not statistically significant.

These results support the value of heterogeneity analysis, revealing that existing financial stress might be a more relevant indicator for how local downturns impact CC students' enrollment choice, while age alone does not necessarily predict a persistence response.

[7] While these specific trends do not survive the stringent Bonferroni correction, they may emerge as robust under a false discovery rate framework. If the analysis were to adopt Benjamini-Hochberg's false discovery rate (FDR) control, which is less conservative and appropriate when some false discoveries are tolerable, additional findings would be deemed significant. However, in all cases examined below, all of the estimates that failed the Bonferroni correction failed the BH thresholds as well.

Table 2

DiD Estimates: Persistence Response to Local Employment Downturns

Persistence	Estimate (pp)	SE		WCB[a] *p*	Bonferroni[b]
A. Binary: Harder-Hit Counties[c]					
Full sample	0.021	0.017		0.108	--
Adult learners	-0.007	0.012		0.559	--
Worked two years pre-college	0.029	0.012	**	0.003	√
Worked one year pre-college	0.039	0.014	**	0.004	√
Full-time starters	-0.078	0.012	***	0.002	√
FAFSA filers	0.084	0.026	***	0.050	X
Pell-eligible	0.175	0.015	***	0.000	√
First-generation	-0.003	0.006		0.592	--
B. Continuous: Dosage[c]					
Full sample	0.007	0.013		0.519	--
Adult learners	0.012	(0.008)		0.153	--
Worked two years pre-college	0.004	(0.009)		0.629	--
Worked one year pre-college	0.004	(0.012)		0.732	--
Full-time starters	-0.004	(0.012)		0.740	--
FAFSA filers	-0.010	(0.018)		0.556	--
Pell-eligible	0.017	(0.008)	**	0.025	X
First-generation	0.002	(0.005)	--	0.647	--

Note. Estimates are from a difference-in-differences (DiD) specification interacting a county-level employment shock with a post-period indicator. All models include individual and year fixed effects. Cluster-robust standard errors at the county level are shown in parentheses. [a] WCB = Wild Cluster Bootstrap. [b] Bonferroni threshold: $p < 0.007$ ($\alpha/7$). [c] Each model is estimated using both a continuous and a binary measure of local downturns. The binary measure of harder-hit counties dichotomizes counties at the median level of predicted downturn. The continuous measure is the absolute magnitude of the predicted employment decline, so that a positive coefficient indicates a larger response to a more severe downturn; descriptive figures retain the original negative sign. Estimates are interaction coefficients, representing differential effect for subgroup relative to complement. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Additionally, a few notable patterns appear in the secondary outcome analysis. Full-time persistence shows generally similar patterns to the binary measure of full-time persistence, with higher probability to persist full-time concentrated among adult students (+7.7%, WCB p<0.007), while Pell-eligible students show minimal difference in their probability to persist full-time versus part-time in college.

Regarding student employment, whether worked while persisting is not impacted by local downturns for the full sample and all focal subpopulations. Although those employed one year before college might work less while persisting (-5.5pp, WCB p>0.007), the p-value suggests

potential association between their work status during college and recession severity, instead of a causal claim. The continuous specification of local downturns suggests a similar pattern for them (-1.3pp, WCB $p<0.007$). Adult students show small positive estimates in continuous specifications (+5pp, WCB $p>0.007$) yet negligible effect in harder-hit counties, although neither of the estimates survive the Bonferroni correction.

Earnings

As Table 3 shows, adult students and Pell-eligible students in severely impacted counties both experience annual earnings losses exceeding $2,000 relative to their counterparts in less affected counties (WCB $p<0.007$). Students with recent work histories show smaller but similar losses: those employed in the two years before enrollment earn roughly $700 less than peers in less affected counties (WCB $p<0.007$). These students have stronger labor market attachment, and their earnings fall when local labor market shrinks, producing a direct income effect that compounds the early credential completion shortfall documented above.

In comparison, full-time starters exhibit an opposite pattern: they earn approximately $2,000 annually more than their peers in impacted counties (WCB $p<0.007$). This mirrors the persistence results, where full-time starters are the only group more likely to stop out in response to local downturns, leaving school and increasing earnings, while other groups remained enrolled and experienced earnings decline. Adult, Pell-eligible, and recently employed students' earnings loss and full-time-starters' earnings gain reflects their different labor market positions: the former groups are more deeply embedded in the local economy and bear its downside, while full-time starters are partially insulated by their enrollment and may relatively benefit from a thinned labor market.

Table 3

DiD Estimates: Annual Earnings Changes on Local Employment Downturns

Earnings Change	Estimate ($2015)	SE		WCB[a] *p*	Bonferroni[b]
A. Binary: Harder-Hit Counties[c]					
Full sample	-487	796.64		0.437	
Adult learners	-2113	217.20	***	0.001	√
Worked two years pre-college	-697	138.79	***	0.003	√
Worked one year pre-college	-1215	389.22	***	0.014	X
Full-time starters	2206	247.68	***	0.000	√
FAFSA filers	-856	368.99	**	0.133	X
Pell-eligible	-2029	223.29	***	0.000	√
First-generation	-301	71.60	***	0.017	X
B. Continuous: Dosage[c]					
Full sample	-364	130.68	***	0.048	X
Adult learners	-288	175.45	***	0.023	X
Worked two years pre-college	5	135.56		0.968	
Worked one year pre-college	211	196.92		0.240	
Full-time starters	129	148.25		0.353	
FAFSA filers	567	188.97	***	0.002	√
Pell-eligible	91	106.96		0.227	
First-generation	-166	76.44	**	0.060	X

Note. Estimates are from a difference-in-differences (DiD) specification interacting a county-level employment shock with a post-period indicator. All models include individual and year fixed effects. Cluster-robust standard errors at the county level are shown in parentheses. [a] WCB = Wild Cluster Bootstrap. [b] Bonferroni threshold: $p < 0.007$ ($\alpha/7$). [c] Each model is estimated using both a continuous and a binary measure of local downturns. The binary measure of harder-hit counties dichotomizes counties at the median level of predicted downturn. The continuous measure is the absolute magnitude of the predicted employment decline, so that a positive coefficient indicates a larger response to a more severe downturn; descriptive figures retain the original negative sign. Estimates are interaction coefficients, representing differential effect for subgroup relative to complement. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Additionally, hourly wages are not significantly impacted by local downturns across specifications and subpopulations, providing no evidence on possibility that selective employment in a contracted economy may leave remaining workers in higher-paying positions.

Exploratory Results

Students enrolled in rural colleges don't show notable difference in the recession effects. For instance, the interaction coefficient on whether persisting in college in the binary model is -0.046pp (SE = 0.027), nominally significant at the 10% level, but wild-cluster bootstrap inference with 9,999 replications yields WCB $p = 0.257$. The differential dose-response for rural students in the continuous DiD three-way interaction is -0.003pp, with wild-bootstrap $p = 0.831$.

Across both specifications, there is no evidence that the recession effect on next-year enrollment differed by rural status.

Counties with different industry compositions generally show minimal differences from full-sample estimates. One exception is that students in counties with higher non-manufacturing shares, where local economies consist of more recession-resilient industries, are 1.9 percentage points less likely to work while enrolled (WCB p=0.000). This provides suggestive evidence that working during college is largely driven by local market disruptions and economic stress.

Summary

Overall, local recession exposure has no significant average effect on persistence or earnings for the full sample of already-enrolled CC students. However, this null average effect masks considerable heterogeneity by students' pre-existing background characteristics, revealing two competing channels.

For financially constrained students (Pell-eligible) and those with prior labor market attachment (one to two years of pre-college work experience), downturns strengthen college persistence. Pell-eligible students experience a 17.5 percentage point increase in next-year persistence, while experienced workers see a 2.9 to 3.9 percentage point increase, with both groups appearing to suffer minimal contemporary earnings losses. Conversely, adult students (aged 25 and older) face significant vulnerabilities, bearing annual earnings losses of approximately $2,100.

In contrast, full-time starters are heavily disrupted by local downturns. For this cohort, persistence drops by 7.8 percentage points while annual earnings rise by roughly $2,200. This inverse relationship suggests a financial constraint response, where students stop out of college to substitute education hours for immediate employment opportunities.

These results suggest that local employment downturns trigger divergent behavioral responses and prompt CC students through various mechanisms, shifting financially constrained students deeper into education as local jobs disappear, while pulling full-time students out of college and into the workforce to cope with financial pressure. Additional outcomes are under analysis.

Discussion

Prior literature establishes that CC enrollment is countercyclical (Foote & Grosz, 2020; Goodman & Winkelmann, 2025) and that individual job displacement modestly affects enrollment and educational investment decisions (Minaya et al., 2023; Ost et al., 2018). Yet no prior work has traced how aggregate employment downturns affect the full educational and labor market trajectories of CC entrants beyond initial enrollment – through enrollment persistence and work behaviors. This research fills that gap by combining county-level Bartik-style employment downturns with individual-level administrative data for students who already enrolled in CCs, revealing that employment downturns affect already-enrolled CC students and entering CC students differently.

Whereas existing literature documents that economic downturns increase overall CC enrollment, this study shows that local employment shocks leave aggregate enrollment nearly unchanged. Further, downturns create negative impact on credential completion, while earnings effects depend strongly on which students and how severe the shock is. The coherence of these results becomes clearer through the lens of heterogeneity. The preliminary findings contribute to a growing body of evidence on how CC students navigate economic disruption and extend prior countercyclical enrollment research in an important direction.

The persistence and earnings findings support the value of applying the lens of opportunity cost and financial constraints simultaneously, but the dominating mechanism differs across student types. Employment downturn reduces the opportunity cost of enrollment, as labor markets contract, making the wage foregone by staying in school lower, while also potentially tightening financial constraints. For students whose enrollment decision is primarily driven by opportunity cost—such as Pell-eligible students and recently-employed students—the opportunity cost channel dominates, and persistence rises. For students who started CC with full-time enrollment, the financial constraint channel dominates, and they exit. This division also shows in the earnings effect, with full-time starters being the only group experiencing earnings gains in harder-hit counties. The aggregate null masks these opposing forces nearly perfectly.

Pell-eligible students are, by definition, among the most financially constrained. When local employment contracts, the opportunity cost of staying enrolled falls sharply relative to dropping out to work. For this group, downturns remove the near-term temptation to exit. This mechanism partially mirrors the one documented by Ost et al. (2018) at the firm level, here observed at the county level for financially constrained students.

Similar persistence advantages through decreasing opportunity costs are observed among recently employed students, although the specific mechanism might differ. Because these students have demonstrated labor force attachment, when local conditions deteriorate, the relative return to continued enrollment increases for them as well. This positive finding is partially consistent with Ost et al.'s (2018) study on the impact of firm-level mass layoffs on working students, which suggests the persistence advantage is only notable on intensive margins (credits attempted), not on extensive margins (probability).

In other words, my findings on county-level recessions are partly consistent with this logic presented in Ost et al.'s (2018) firm-level layoff study, but the mechanism appears to operate primarily through equity-differentiated channels: strong positive persistence responses is observed not only among recently employed students, but also, and even more substantially, among low-income students (Pell-eligible). This suggests that for already-enrolled CC students, the opportunity cost mechanism is strongest among students for whom market wages are low and for whom additional time in school carries meaningful career value.

Notably, none of my findings directly support the enrollment-employment substitution that human capital framework predicts. The overall pattern, estimated effects on persistence probability generally exceed those on working, suggests that individuals' enrollment adjustment does not necessarily translate into corresponding changes in working behaviors. Pell-eligible students' persistence probability increases considerably, yet their employment probability and earnings estimates do not change significantly. The pattern reverses for full-time students, whose estimated persistence probability falls as earnings rise, but their employment probability does not change significantly, indicating adjustment on the intensive rather than the extensive margin. Earnings findings reveal a stark contrast between full-time starters and students with stronger prior labor market attachment, including adult students, who suffer concentrated earnings losses with minimal changes in enrollment probability.

The contrast between the year-to-year persistence and early completion findings produces the central paradox of this study. Employment downturns pull economically vulnerable students into continued enrollment, yet they weaken the early completion pipeline. This partially aligns with Goodman and Winkelmann's (2025) finding that the marginal non-enrollee, whose enrollment decisions respond most to labor market conditions, would not have completed a

credential: recession may draw low-commitment students into college, whose attachment to college credentials is weak to begin with, so that enrollment and completion gains need not move together. Because their analysis is institution-level and descriptive, the linked student records used here offer a more direct test of this compositional account.

Nevertheless, because this study can only track early, short-term credential completion status, this pattern does not represent eventual completion. Students may stop out and re-enroll later beyond the observation window, and such enrollment trajectories are common among CC students. Several other non-exclusive mechanisms could produce this pattern during the Great Recession. First, students who persist through economic downturns may shift from credit-bearing courses to non-credit programs, which account for a large share of CC and CTE enrollment (Van Noy et al., 2008; Xu et al., 2023), especially with the TAACCCT programs expanding non-credit, alternative pathways to career training opportunities following the Great Recession (U.S. Department of Labor, n.d.). As this study uses credit-bearing enrollment data, such a shift would not be captured in the analysis and deserve future work. Second, students may lose the momentum associated with completing a college credential (Jenkins & Bailey, 2017), redirecting toward training that does not necessarily lead to a credential. Lastly, institutional capacity in severely impacted counties may have been constrained, limiting academic resources necessary to credential completion.

Taken together, these findings suggest that CCs serving economically vulnerable populations face a distinctive challenge during local downturns. The concentration of prolonged persistence with minimal completion and earnings effects among financially vulnerable students indicates that economic shocks may widen existing inequalities. If confirmed, those findings would point toward the need for recession-responsive financial aid and advising interventions

targeted at students with the least economic cushion. Anticipating this dynamic may be among the more consequential lessons for policy and practice. Strengthening advising infrastructure during economic downturns, with particular attention to helping students make durable program choices amid uncertainty, may be as important as expanding program capacity for new enrollees.

Next Steps and Study Limitations

Estimates for the continuous measure will be standardized in subsequent analyses to facilitate effect size comparisons across outcomes.

Ongoing analyses continue to refine these estimates for the primary outcomes and secondary outcomes, which supplements the narrative on potential mechanisms discussed above. Estimates for the continuous measure will be standardized in subsequent analyses to facilitate effect size comparisons across outcomes.

Secondary outcomes, particularly program choice, offer further purchase on mechanisms. Students may concentrate in some CTE programs with strong local labor market returns which do not necessarily lead to credential completion (e.g., Bahr et al., 2023). While most prior research suggests completion effects are generally modest but vary across fields, with shorter-term or high-return CTE fields behaving differently, so field-level analysis is exploratory yet informative. That said, declared major data has limitations as a measure of program of study, especially among CC students, for whom defining CTE by declared major versus by attempted coursework markedly change who are count as CTE students (Yu et al., 2026). While course data is not available here, future work using course-taking records could establish program choice more directly.

This analysis estimates college outcomes using contemporaneous downturn exposure (the Bartik measure of the Minnesota county where the student is enrolled). In order to explore

longer-term outcomes, such as completion and earnings outcome at a fixed number of years post-college, I need to use alternative, predetermined downturn (the Bartik measure of the first Minnesota community college county) to minimize endogeneity issues that later county of residence is potentially endogenous to the outcomes themselves, exploiting the to study the post-college earnings outcomes. Mobility in the sample supports this: 62% of CC starters remain in their original county, so the initial exposure closely approximates contemporaneous exposure for most students while avoiding bias from endogenous moves.

As a specification check, I constructed a placebo shift-share instrument from industries with minimal national employment decline during the Great Recession. Results from this test are forthcoming.

Finally, with 35 counties the analysis is underpowered for examining how institutional and county characteristics condition student responses, including local industry composition, rurality, and institutional capacity. These moderators are substantively important, and larger geographic samples with richer variation would allow causal analysis of them.

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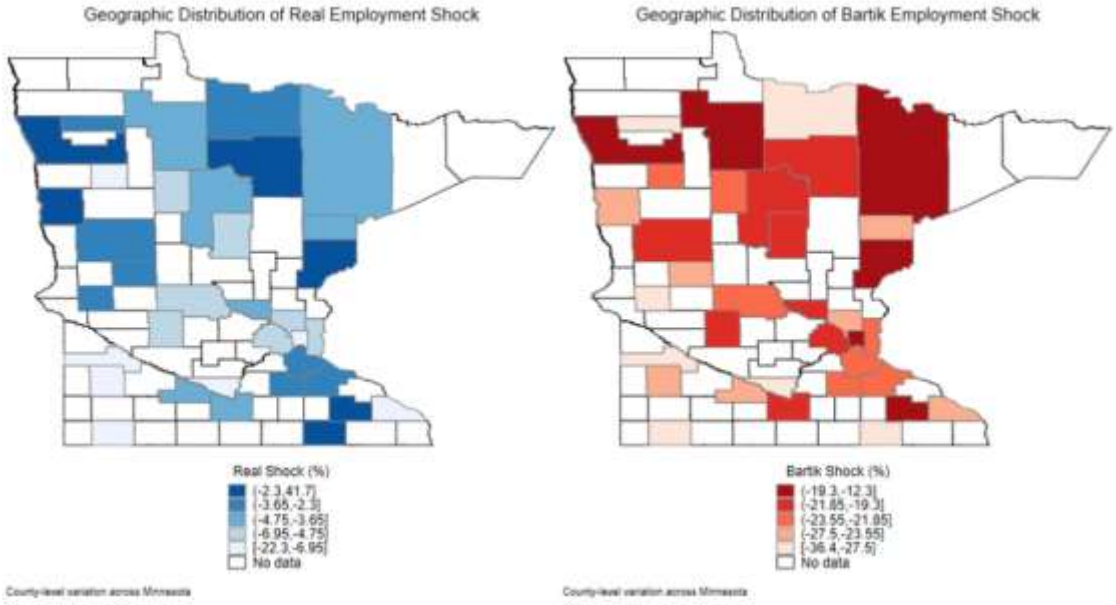
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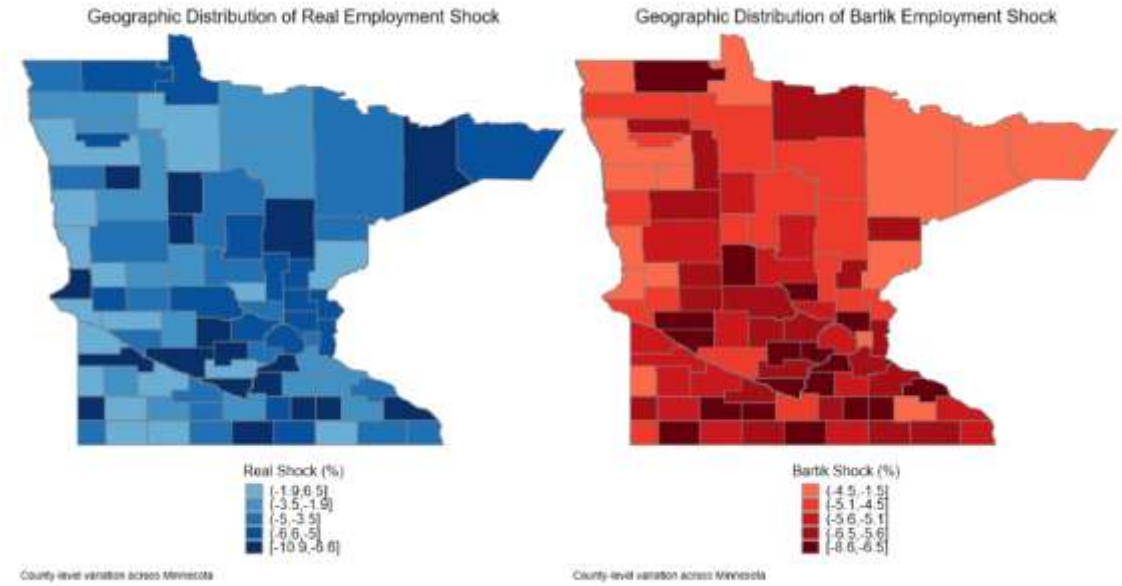
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Appendix



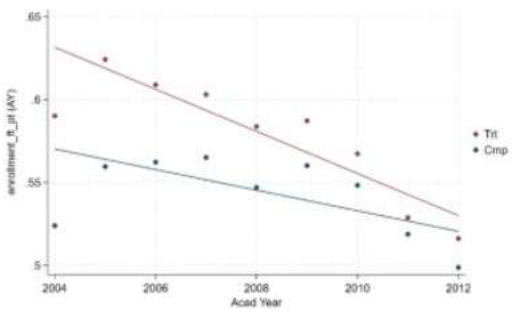
(a) Sampled counties



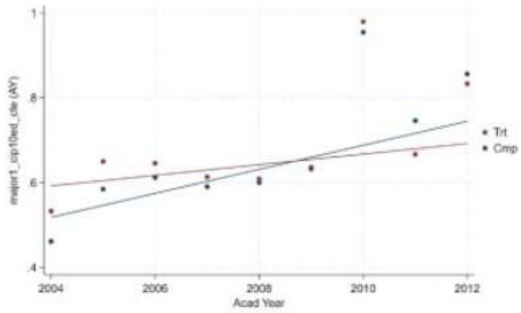
(b) All counties

Figure A1
Geographical Distribution of Real and Bartik Employment Shocks

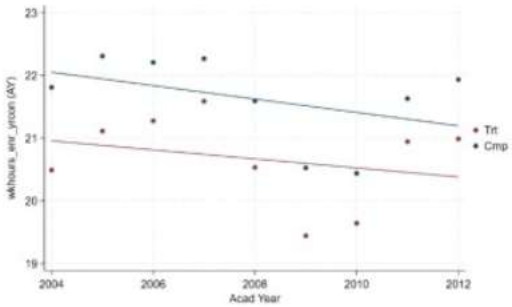
Note. Recession measures are expressed as their original negative values (i.e., proportional employment declines) to preserve intuitive directionality in descriptive displays.



Full-time enrollment



CTE majors



Weekly hours worked during college

Figure A2

Longitudinal Trend in Student Enrollment and Work Behaviors by Recession Severity: Treatment and Comparison Counties

Note. Counties are classified into treatment and comparison groups by dichotomizing predicted downturns at the median, where the treatment group ("Trt") comprises counties experiencing above-median employment declines and the comparison group ("Cmp") comprises counties experiencing below-median declines. Trends are displayed separately to illustrate student trajectories across groups.